



4 The continuous random variable X has probability density function f given by

$$f(x) = \begin{cases} kx^3 & 0 \leq x < 1, \\ k(5-x) & 1 \leq x \leq 5, \\ 0 & \text{otherwise,} \end{cases}$$

where k is a constant.

(a) Sketch the graph of f .

[1]

(b) Show that $k = \frac{4}{33}$.

[2]

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(c) Find the cumulative distribution function of X . [3]

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(d) Find the median value of X . [4]

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